

Root Words Cheat Sheet II — Scale, Structure & Mathematical Physics

This sheet covers two different kinds of root: the SI scale prefixes, which form a genuine ordered sequence from small to large, and the conceptual roots of mathematical physics, which form families rather than a ladder. The scale ladder comes first, laid out as a sequence, since order carries real information there. Everything after it is grouped by idea, exactly as on the first sheet.

The Scale Ladder

Prefix	Factor	Root meaning	Origin
nano-	10^{-9}	dwarf	Greek <i>nanos</i>
micro-	10^{-6}	small	Greek <i>mikros</i>
milli-	10^{-3}	thousand(th)	Latin <i>mille</i>
centi-	10^{-2}	hundred(th)	Latin <i>centum</i>
deci-	10^{-1}	ten(th)	Latin <i>decem</i>
— <i>base unit</i> —			
deca-	10^1	ten	Greek <i>deka</i>
hecto-	10^2	hundred	Greek <i>hekatón</i>
kilo-	10^3	thousand	Greek <i>khilioi</i>
mega-	10^6	great	Greek <i>megas</i>
giga-	10^9	giant	Greek <i>gigas</i>
tera-	10^{12}	monster	Greek <i>teras</i>

Notice that deci- and deca- both come from "ten" but pull in opposite directions, one Latin, one Greek, which is exactly why they get confused. Notice too that kilo- upward are Greek words for bigness (great, giant, monster) repurposed as clean multipliers, not literal counting words the way deca- and hecto- are.

Notice also that per-cent, % = *c* follows from "per" (for each) and "cent" latin **centum** for hundred.

PER- *through, for each* (Latin *per*)

percent, per second, per unit, permeability, permittivity, periodic

CENT-/CENTI- *hundred* (Latin *centum*)

percent, centimetre, centigrade, century

Trap: centripetal is NOT this root — it's centr- (centre), a false friend.

The Morphism Triad — MORPH- *form, shape* (Greek *morphe*)

isomorphism (iso- = same form), **homomorphism** (homo- = same form, weaker sense), **polymorphism** (poly- = many forms), plus metamorphosis, amorphous.

The etymology here is not decoration: it *is* the mathematical definition. A structure-preserving map between two objects is named "same form" because that is precisely what it does.

Conic Sections — **PARA-/HYPER-/ELLIPS-** (Greek, all sharing *-bola/-leipsis*, "a throw")

parabola = para- (alongside) + ballein (to throw): a throwing alongside.

hyperbola = hyper- (beyond, excess) + throw: a throwing beyond.

ellipse = elleipsis: a falling short.

How a thrown line lands relative to a reference: exactly alongside, overshooting, or falling short.

INTEGR- *whole, untouched* (Latin *integer*)

integer, integral, integrate

FRACT-/FRAG- *break* (Latin *frangere*)

fraction, fracture, fragment, fractal, diffraction, refraction

RATIO-/RATIONAL- *reckoning, comparison* (Latin *ratio*)

ratio, rational, irrational, proportional

NUMER-/DENOM- *number / name* (Latin *numerus / nomen*)

numerator, numerical, enumerate, denominator

-NOMIAL *named term* (Latin *nomen*)

monomial, binomial, polynomial, nomenclature

BI- *twice* (Latin *bis*)

bilinear, binary, bisect, binomial

DERIV- *flowing from* (Latin *de- + rivus*, stream)

derivative, derive

Same root as "river".

RECIPRO- *back and forth* (Latin *reciprocus*)

reciprocal

EXPO-/EXPONENT- *putting out, displaying* (Latin *exponere*)

exponent, exponential, exposure

LOG-/LOGO- *word, reckoning, ratio* (Greek *logos*)

logarithm, logic

"Logarithm" = logos + arithmos, literally "ratio-number."

POLY- *many* (Greek *polys*)

polynomial, polygon, polymer

QUAD-/QUART- *four* (Latin *quattuor*)

quadratic, quadrant, quarter, quadrupole

Test Yourself

1. Order these five prefixes from smallest to largest factor: *kilo-*, *nano-*, *centi-*, *mega-*, *milli-*.
2. Explain why *deci-* and *deca-* are easy to confuse, and state which one shrinks and which one grows.
3. *Integer* and *fraction* come from roots with opposite meanings. State both roots and their meanings.
4. Break down *polynomial* into its two roots and explain how the literal meaning matches the mathematical object.
5. *Derivative* shares a root with an everyday English word for a flowing body of water. Name the word and the shared root.
6. Explain, using its roots, why "logarithm" literally means "ratio-number."
7. *Isomorphism*, *homomorphism*, and *polymorphism* all end in *-morphism*. Explain why "same form" (or "many forms") is a mathematically accurate name for what each of these objects does, not just a label.
8. *Parabola*, *hyperbola*, and *ellipse* are all named from the same throwing image. Explain what "throwing beyond" tells you about the shape of a hyperbola compared to a parabola.
9. *Centripetal* looks like it should share a root with *percent* and *centimetre*, but it doesn't. Explain the trap and give the correct root.
10. *Reciprocal* comes from a root meaning "back and forth." Explain why this is an apt name for the operation $x \mapsto \frac{1}{x}$.

Answer Bank

1. Smallest to largest: nano- (10^{-9}), milli- (10^{-3}), centi- (10^{-2}), kilo- (10^3), mega- (10^6).
2. Both mean "ten," but *deci-* is Latin and shrinks (10^{-1}), while *deca-* is Greek and grows (10^1). The shared meaning with opposite direction is exactly what makes them confusable.
3. *Integer* from *integer* (Latin, "whole, untouched"); *fraction* from *frangere* (Latin, "to break"). One names wholeness, the other names the act of breaking a whole into parts.
4. *Poly-* (many) + *-nomial* (named term, from *nomen*): literally "many-termed," which is exactly what a polynomial is, an expression built from several named terms added together.
5. *River*, from the same root *rivus* (stream). A derivative "flows from" a function in the same sense a river flows from its source.
6. *Logos* (Greek, word/reckoning/ratio) + *arithmos* (Greek, number): a logarithm is literally a "ratio-number," reflecting its original use in comparing ratios of quantities.
7. Each word names a map between two mathematical objects, and the ending correctly describes what kind of map it is: an isomorphism preserves structure exactly (same form both ways), a homomorphism preserves structure in one direction, and polymorphism describes one interface taking many forms. The name is not a metaphor bolted on afterward; it states the mathematical property directly.
8. "Throwing beyond" means the curve's associated line overshoots the reference rather than landing exactly alongside it (parabola) or falling short of it (ellipse). This maps onto the fact that a hyperbola has two open, ever-widening branches rather than the single closed curve of an ellipse or the single open curve of a parabola.
9. The trap is that *cent-* (hundred) and *centr-* (centre) look almost identical but are unrelated roots. *Centripetal* is built from *centr-* + *petere* ("to seek the centre"), nothing to do with the number hundred.
10. *Reciprocus* describes something that moves back and forth between two states. Taking the reciprocal of x and then taking the reciprocal again returns you to x , a genuine back-and-forth, which is exactly the operation's behaviour.

Matching Quiz: Prefix to Meaning

Draw a line (or write the matching letter) connecting each prefix/root on the left to its correct meaning on the right. Meanings are deliberately shuffled and there are more meanings than prefixes, so some will be left over.

Prefix/Root		Meaning
1. NANO-	_____	A. break
2. INTEGR-	_____	B. dwarf
3. FRACT-	_____	C. whole, untouched
4. POLY-	_____	D. back and forth
5. MORPH-	_____	E. many
6. RECIPRO-	_____	F. form, shape
7. DERIV-	_____	G. four
8. LOG-/LOGO-	_____	H. flowing from
9. QUAD-	_____	I. hundred
10. PARA-	_____	J. word, ratio
		K. alongside
		L. ten

Bonus: pick one leftover meaning from the right-hand column and name a word from either sheet that uses it.

Tricky Words: Construct Your Own Definition

1. **Rationalise** (*ratio* = reckoning, comparison, *-ise* = to make/turn into)
Your definition:

2. **Bijection** (*bi-* = twice, *-jection* from *jacere* = to throw)
Your definition:

3. **Congruent** (*con-* = together, *gruere* = to agree, come together)
Your definition:

4. **Asymptote** (*a-* = not, *syn-* = together, *ptotos* = falling)
Your definition:

5. **Coefficient** (*co-* = together, *efficere* = to bring about, effect)
Your definition:

6. **Differential** (*dis-* = apart, *ferre* = to carry/bear)
Your definition: